BorNEO HackWknd 2026

MoneyMate Project Report

SUBMITTED BY

(Team DEDSEC)

Daryl Harry Bennet

Fivryl Anak Wilson

Nur Izzati Binti Haudi

Nurkhairunnisa Nazihah Binti Mohd Fadlie

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INTRODUCTION

AI for financial literacy

Digital financial services in ASEAN are growing swiftly, with digital payments expected to surpass $1.5 trillion by 2030, yet financial literacy rates remain worryingly low. A notable discrepancy exists where the availability of financial instruments (such as e wallets and Buy Now, Pay Later options) surpasses the user's comprehension of risk, interest, and managing debt. The ASEAN area encounters a notable protection gap, as financial losses from disasters frequently remain uninsured due to insufficient financial resilience. Additionally, numerous people find it challenging to comprehend intricate financial services, restricting their access to insurance and investment options.

AI is making financial education more accessible with robo-advisors and customized budgeting apps that offer round-the-clock guidance aligned with an individual's objectives, These AI-based applications utilize real-time data to monitor expenditure trends and provide instant, actionable advice for saving. For students and entrepreneurs, AI streamlines financial management, aiding them in grasping the long-term effects of their financial choices. Addressing these knowledge gaps is crucial for creating a financially robust ASEAN that can endure economic and climate-related challenges.

Chosen SDG

I) SDG 9: Industry, Innovation & Infrastructure

SDG 9 focuses on developing technological innovation to bridge digital gaps. Our project, MoneyMate, directly contributes to this goal by creating an innovative financial management platform that leverages AI to build personal financial infrastructure. By democratizing access to financial tools and education, we help bridge the digital divide in financial literacy across ASEAN communities.

II) Problem Statement

Despite the rapid digitalization of financial services across ASEAN, a critical disconnect persists between access and understanding. Young adults, students, and entrepreneurs actively participate in the digital economy through e-wallets, online banking, and buy now pay later schemes. However, they lack a centralized platform that not only tracks their finances but actively educates them on financial health. This gap leaves many individuals vulnerable to debt mismanagement, unprotected against economic shocks, and excluded from wealth-building opportunities simply because they were never taught how to navigate an increasingly complex financial landscape.

III) Objectives

- **Simplify Financial Tracking:** Provide users with an intuitive dashboard to monitor income, expenses, and savings in one place
- **Enhance Financial Literacy through AI:** Lay the foundation for personalized financial insights and educational tools that help users understand their financial behavior
- **Promote Goal-Oriented Saving:** Enable users to set, visualize, and track progress toward financial goals through engaging progress indicators
- **Centralize Financial Actions:** Consolidate expense tracking, bill payments, budget management, and reporting into a unified platform

REVIEW OF EXISTING SOLUTIONS

Current Market Solutions

- **Banking Apps (Maybank2U, CIMB Clicks):** Offer basic transaction history and spending categorization within their banking ecosystems

- **Budgeting Apps (Money Lover, Spendee):** Focus on expense tracking and budget setting, with premium tiers for additional features

- **E-Wallet Platforms (Touch 'n Go, GrabPay):** Provide transaction histories limited to their own platforms rather than holistic financial views

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- **Investment Platforms (StashAway, Wahed):** Specialize in portfolio management and robo-advising

Identified Gaps

- **Fragmented Financial Picture:** Users must juggle multiple apps to see their complete financial picture across banks, e-wallets, and investments

- **Limited Educational Value:** Existing tools prioritize recording transactions over teaching users how to make better financial decisions

- **Generic User Experience:** Most financial tools use a one-size-fits-all approach that doesn't account for different levels of financial knowledge, which ends up pushing away the very people who need guidance the most

- **Reactive Design:** Most apps simply log past spending rather than proactively guiding future behaviour

- **Superficial AI Implementation:** Where AI exists, it is typically limited to automated categorization rather than meaningful financial coaching

What MoneyMate aims to Fix

MoneyMate addresses these gaps by consolidating financial data into one unified platform, prioritizing education alongside tracking, offering personalized insights, providing proactive nudges, designing for accessibility, and building a foundation for true AI-powered financial coaching.

METHODOLOGY

Technical Track

MoneyMate was built using a front-end focused tech stack that prioritizes rapid development, clean design, and seamless user experience.

I) Tech Stack

- **HTML5:** Structured the semantic layout of all web pages, ensuring accessibility, and proper document flow
- **CSS3:** Implemented custom styling with CSS variables for dynamic theming, flexbox and grid for responsive layouts, and transitions for smooth interactions
- **JavaScript:** Powered all client-side functionality including authentication, modal management, form validation, and local storage operations
- **Local Storage API:** Enabled user data persistence across sessions without requiring backend infrastructure

II) Architecture

Our application follows modular front-end architecture with clear separation of concerns such as:

- **Component Based Structure:** Reusable components including modals, cards, and form elements maintain consistency across the application while reducing code duplication
- **Centralized Styling:** All styles are managed through a single CSS file with CSS variables enabling seamless theme switching and easy maintenance
- **Client-Side State Management:** Local storage handles user sessions, authentication state, and data persistence without complex backend requirements
- **Event-Driven Interactions:** JavaScript event listeners manage user interactions, modal toggles, form submissions, and real-time UI updates

III) Implementation Process

Phase 1: Planning and Design

- Analyzed financial literacy gaps in ASEAN based on provided case study materials
- Mapped user journeys and identified core features needed for an effective financial companion
- Defined colour schemes and dark/light mode variables for accessibility

Phase 2: Core Structure

- Established project folder structure with separate HTML, CSS< and JavaScript files
- Built the foundational layout including header controls, containers, and navigation elements

- Implemented the theme toggle system with localStorage persistence
- Created reusable modal components for expenses, payments, goals, and budgets

Phase 3: Authentication Implementation

- Developed the login/register interface with animated toggle slider between modes
- Implemented form validation for password matching and required fields
- Set up localStorage to store user credentials and manage session state
- Created the authentication flow from landing page to dashboard access

Phase 4: Dashboard Development

- Built the main dashboard layout with sidebar navigation and content sections
- Implemented section switching functionality to display different views such as Home, Expenses, Payments, Profile, and Reports
- Developed interactive components

Phase 5: Interactive Features

- Created the three-step payment method simulating real banking transactions
- Implemented OTP verification flow for security demonstration
- Built budget management interface with real-time calculation of allocations
- Developed goal-setting functionality with form validation
- Developed chatbot providing financial guidance
- Added expandable sections in the profile for settings management

Phase 6: Testing and Refinement

- Conducted internal testing across different user flows
- Refined UI interactions and transition animations
- Ensured theme switching applied consistently across all components
- Validated form validation and error message display

RESULTS

Summary of Outcomes

The development of the MoneyMate prototype successfully translated our proposed solution from concept into a tangible, interactive user interface and we achieved our core objective of creating a non-functional prototype that visually demonstrates how AI can bridge the financial literacy gap in ASEAN.

Objectives Achieved

- **Simplify Financial Tracking:** The dashboard provides a clean, centralized view of total balance, savings, and spending. Recent expenses and activity logs demonstrate intuitive tracking capabilities that make financial monitoring effortless.

- **Enhance Financial Literacy through AI:** While full AI integration is pending further development, the foundation is laid through smart budgeting tools, goal-setting features, and financial reports that educate users on their spending patterns and promote better financial decision-making.

- **Promote Goal-Oriented Saving:** The Goals section with visual progress bars and the "Add Goal" functionality allows users to set and track savings targets, transforming abstract financial planning into a tangible and engaging experience.

- **Centralize Financial Actions:** The prototype successfully consolidates multiple financial functions such as expense tracking, bill payments, budget management, and report generation, into a single, unified platform that eliminates the need for multiple financial apps.

Key Deliverables

- Complete user authentication flow with login and registration using local storage
- Five-section dashboard including Home, Expenses, Payments, Profile, and Reports. This showcases comprehensive financial management
- Goal-setting functionality with visual progress tracking
- Notification toggle for useful reminders
- Chatbot for financial guidance
- Dark/light mode theme toggle for enhanced user experience
- Responsive modal components for expenses, payments, goals, and budgets

PROTOTYPE SHOWCASE

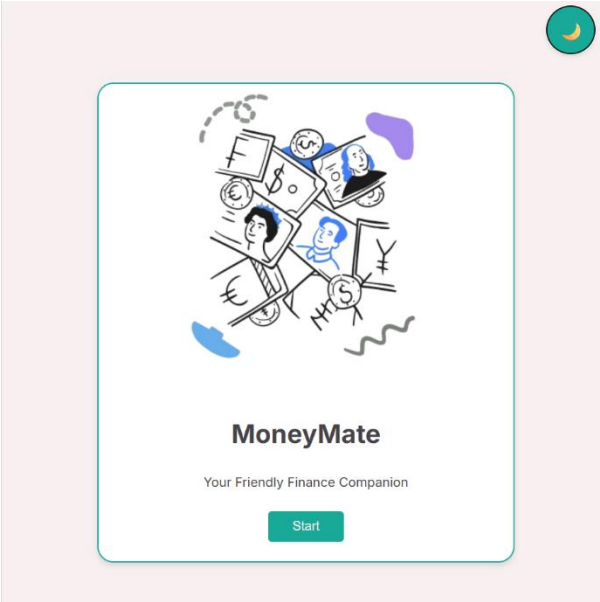
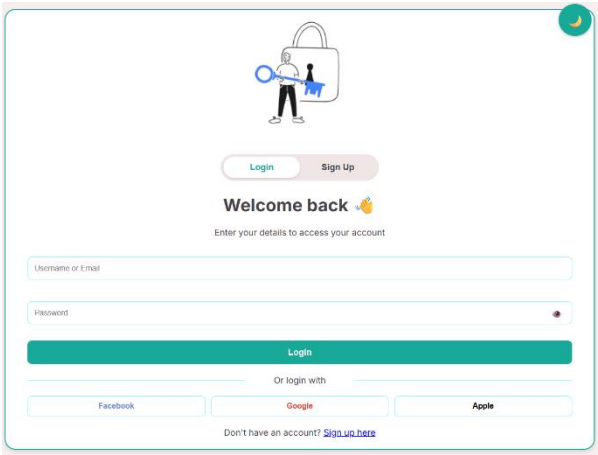


Figure 1 - Landing Page

The entry point to MoneyMate features a clean, welcoming interface with the app's logo and a prominent Start button. The fixed header controls include the theme toggle for dark/light mode access from any page.



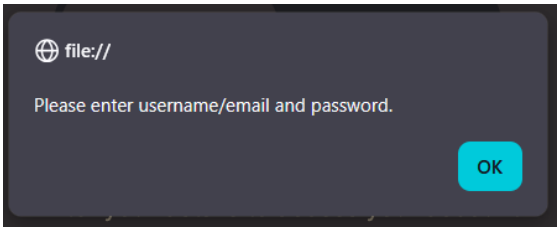


Figure 2 - Login Interface

Users can securely access their accounts using either username or email. The password field includes a toggle to show/hide characters, and the interface provides clear feedback for invalid credentials.

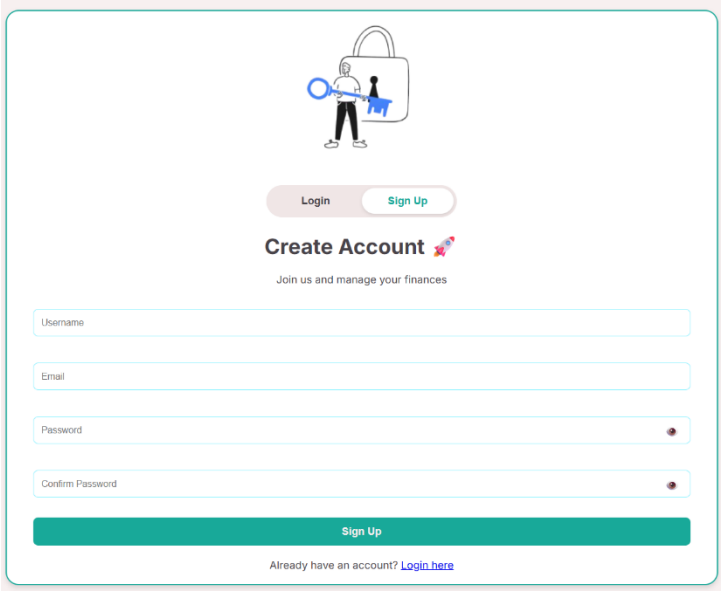


Figure 3 - Sign Up Interface

New users can create accounts with username, email, and password confirmation. Form validation ensures passwords match and all fields are completed before submission.

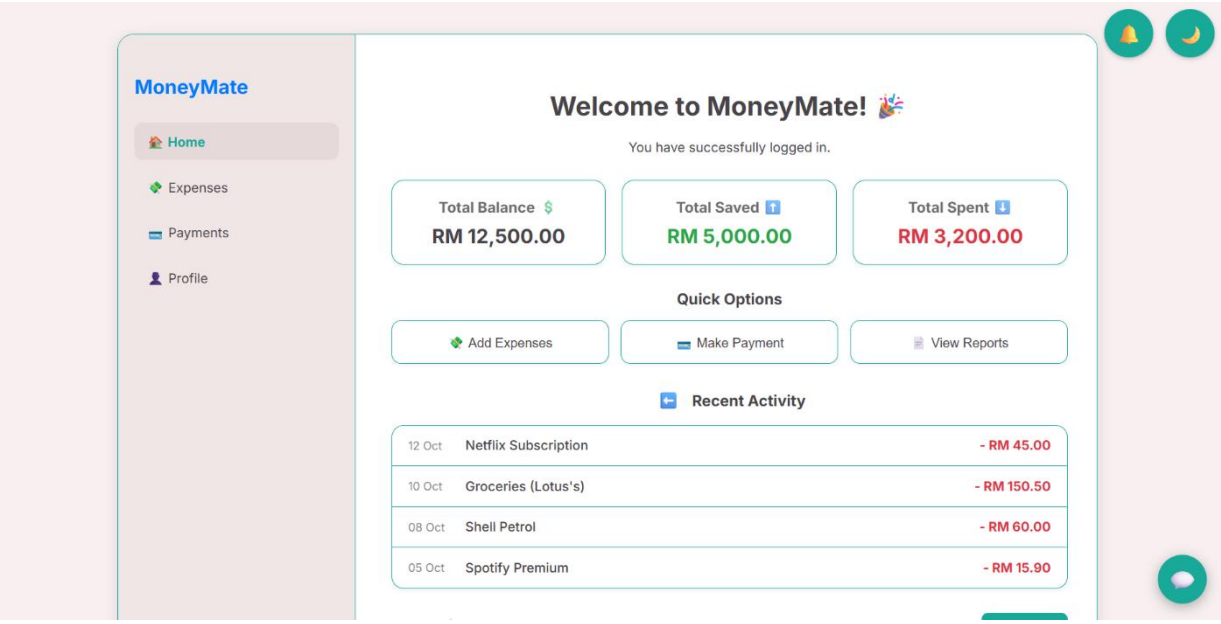


Figure 4 – Dashboard

Upon successful login, users are greeted with a comprehensive dashboard displaying their financial overview. The Home section features total balance, savings, and spending windgets, along with quick option buttons for common tasks, as well as recent activity feed showing the user's latest transactions.

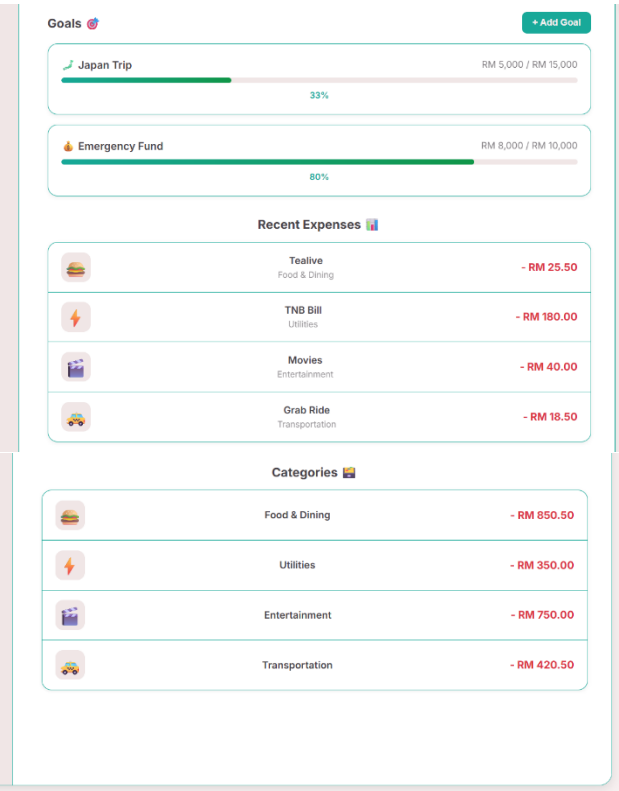


Figure 5 – Scrolled Down Dashboard

As the user scrolls down the Home menu, they will be met with Goals Progress Trackers, where visual progress bars are shown along with savings goals and their current statuses. There I also the Recent Expenses list with a categorized display of recent spendings with icons.

At the end of the Home menu, there is the Categories overview displaying a summary of spending by category showing totals of spending by the user.

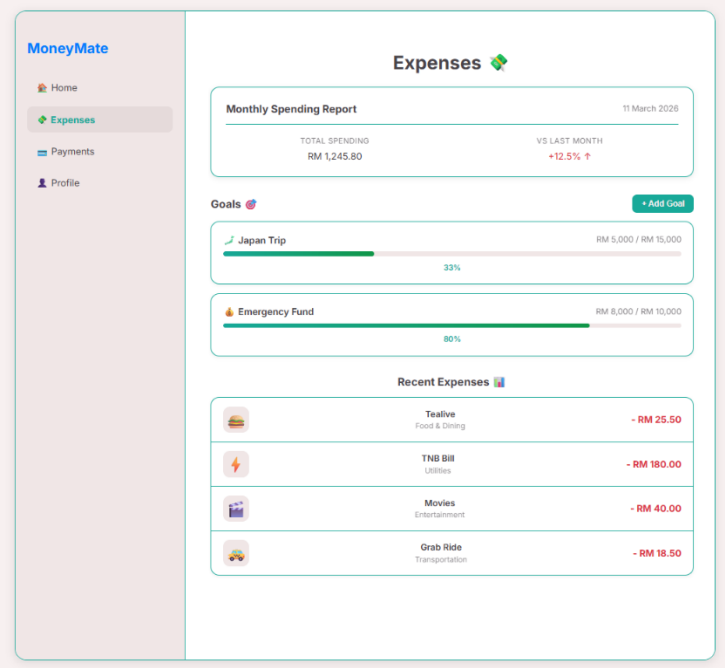


Figure 6 – Expenses Section

The expenses section provides detailed spending reports with monthly comparisons. Users can view their spending patterns and track category-specific expenses.

The Monthly Spending Report Card displays total spending for the current month (RM1,245.80) alongside a comparison to the previous month showing a 12.5% increase. The goals section and recent expenses list are synchronized with the home dashboard to maintain consistency across views.

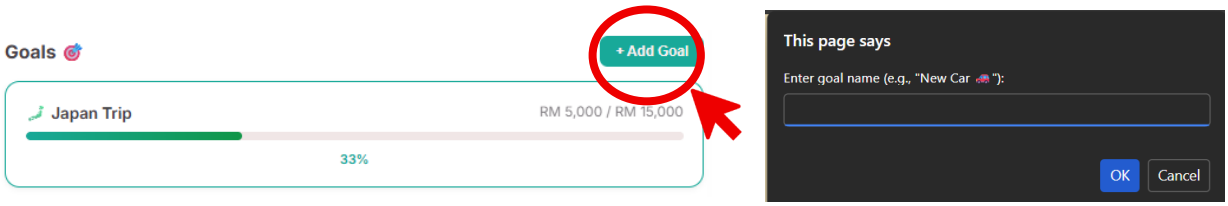


Figure 7 - Goals Modal

Goals can be added from the "Add Goal" button. This process functions as a quick option button. A pop-up will be displayed where users can input their new goals

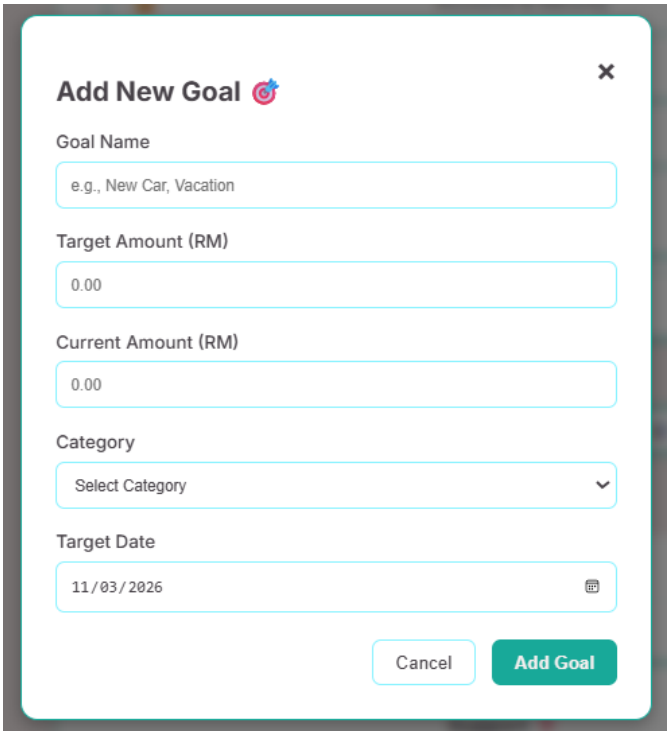


Figure 8 - Goals Modal (Financial Tools Menu)

Alternatively, new goals can also be added through the profile section. This is the full goal management interface where users can:

- Create new savings goals with specific target amounts and deadlines
- Edit or delete existing goals
- Categorize goals by type (Travel, Emergency Fund, Education, Investment, Shopping, Other)
- View detailed breakdowns of all active goals in one place
- Access historical data on completed goals
- Configure goal specific settings and notifications

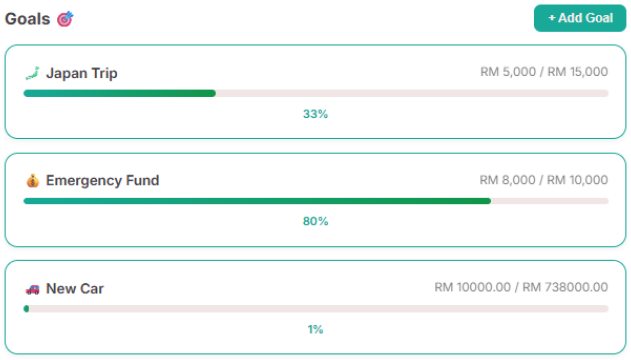


Figure 9 - New Goal Added

Users can create and track financial goals with target amounts and deadlines. Visual progress bars provide immediate feedback on savings progress.

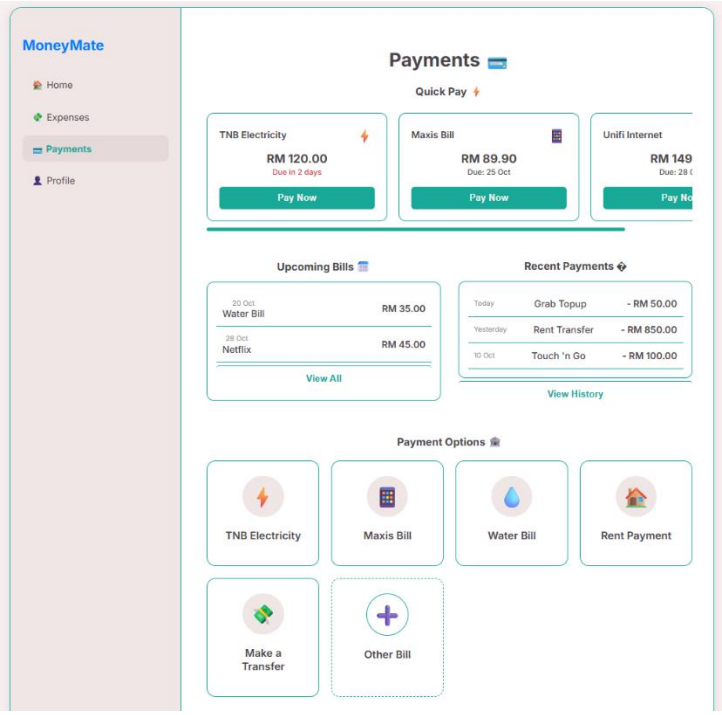


Figure 10 - Payments Section

The payments hub offers multiple ways to manage bills and transfers.

Quick Pay Cards display pending bills with its key details.

Upcoming Bills List shows additional pending payments with a "View All" to expand the list.

Recent Payments History tracks completed transactions with their timestamps.

Payment Options Grid presents clickable cards for common billers and an option for adding new payees.

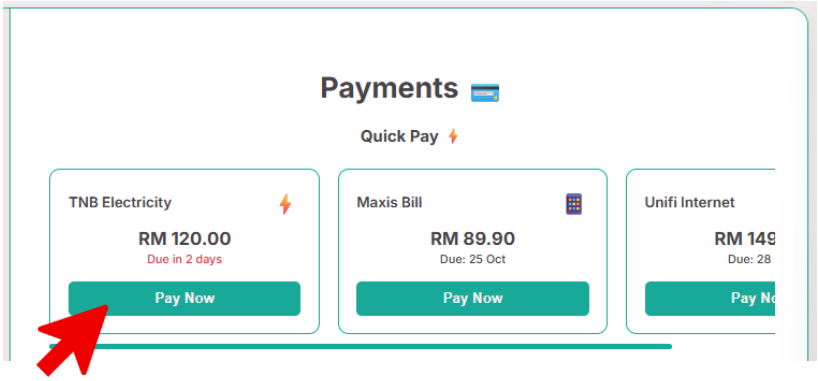


Figure 11 - Payment Modal Flow

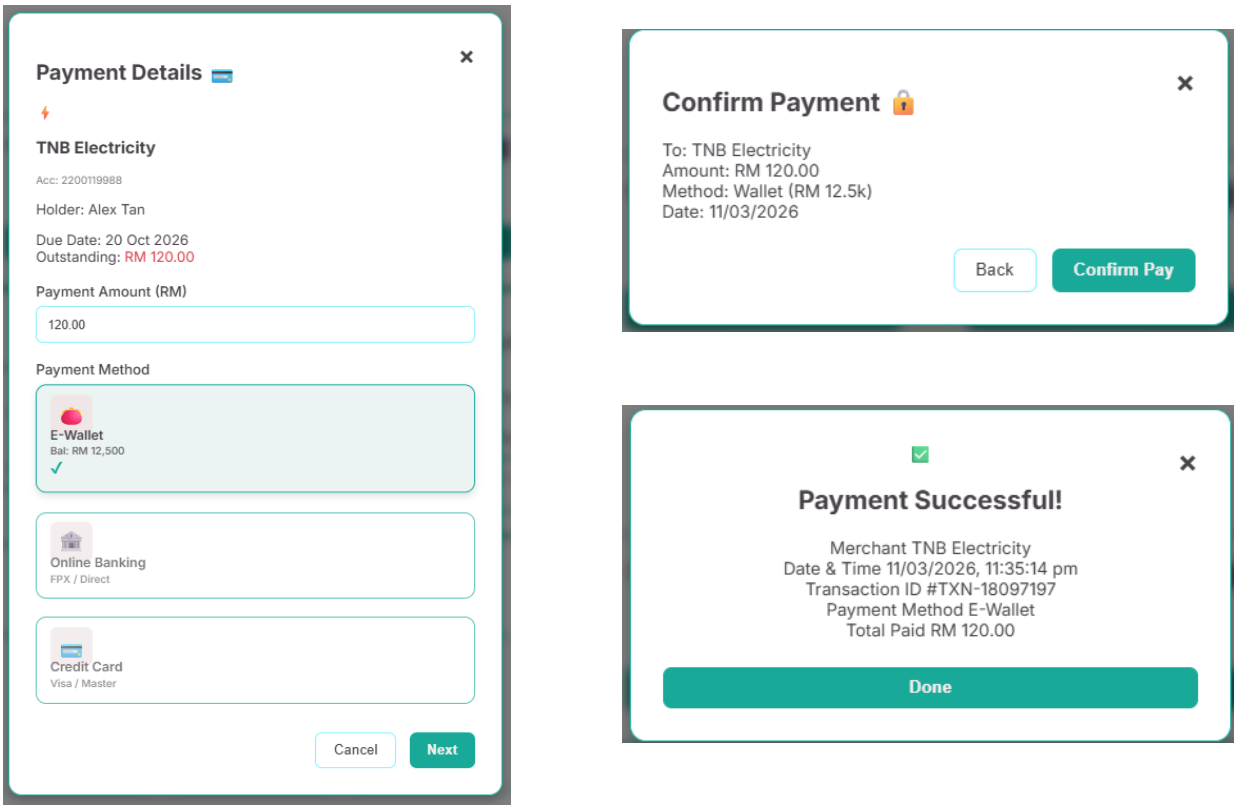


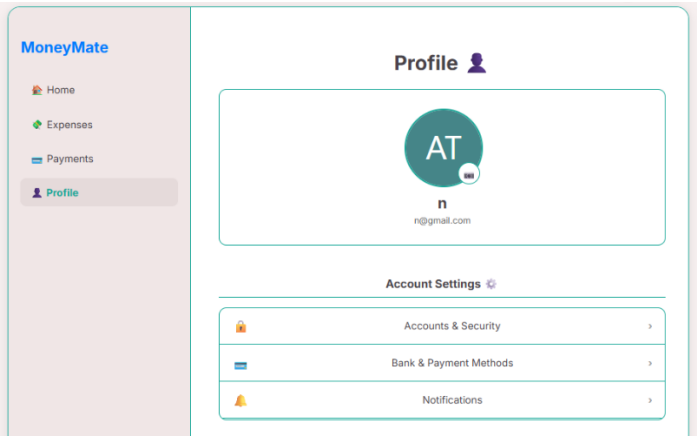
Figure 12 - Payment Modal Step 2 & 3

The three-step payment process simulates a real banking experience, guiding users through details entry, confirmation, and receipt generation.

Step 1 shows payment details with biller information, amount entry, and payment method selection.

Step 2 confirmation screen showing payment summary for user verification before finalizing.

Step 3 displays success receipts with transaction ID, date/time, and payment confirmation details.



The profile section centralizes account management, security settings and financial tools. Users can update their profile image, manage connected accounts, and access support resources.

Figure 13 - Profile Section

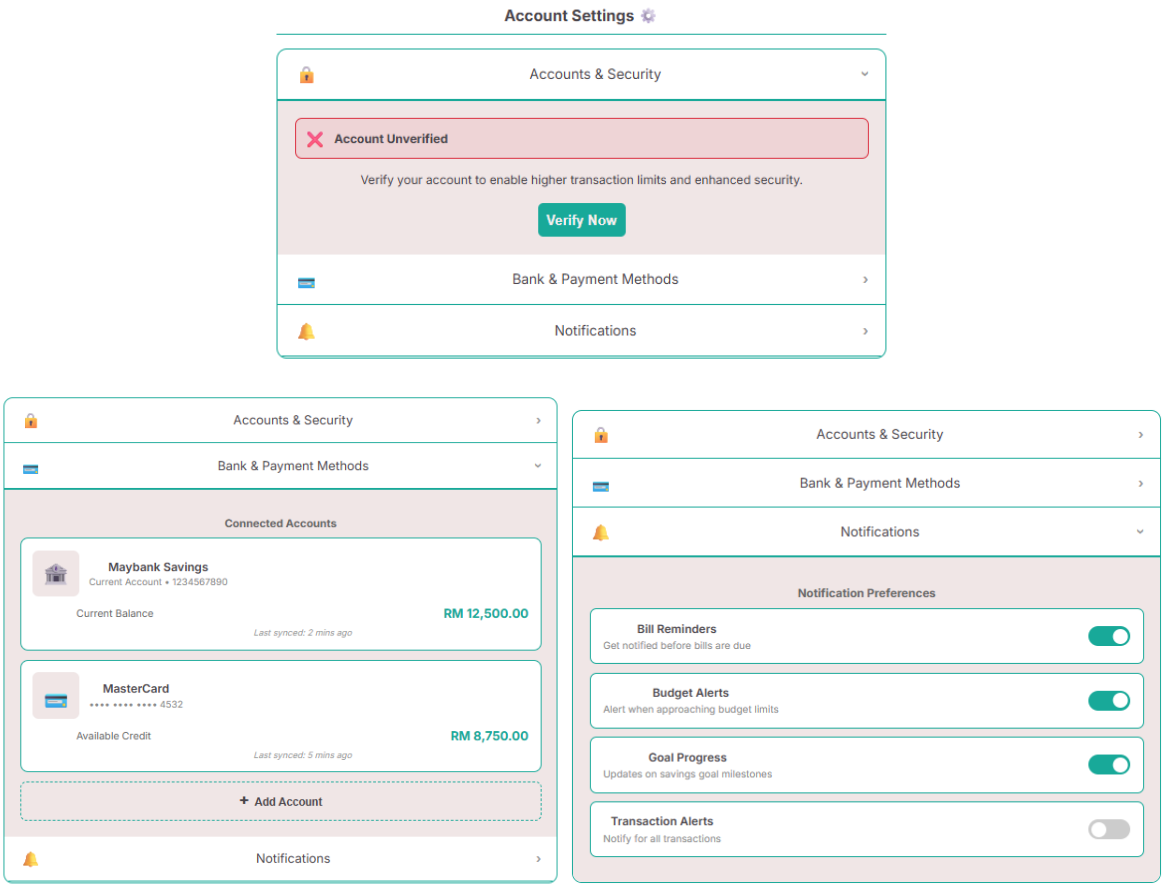


Figure 14 - Account Settings Menu Expanded

Account Settings expandable sections include:

- Accounts & Security with OTP verification flow for account validation
- Bank & Payment Methods showing connected Maybank savings account and MasterCard with current balances
- Notifications with toggle switches for bill reminders, budget alerts, goal progress, and transaction alerts

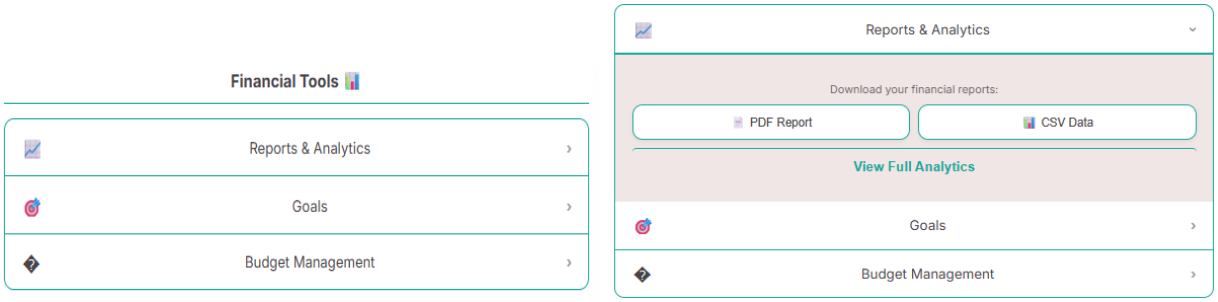


Figure 15 - Financial Tools Menu

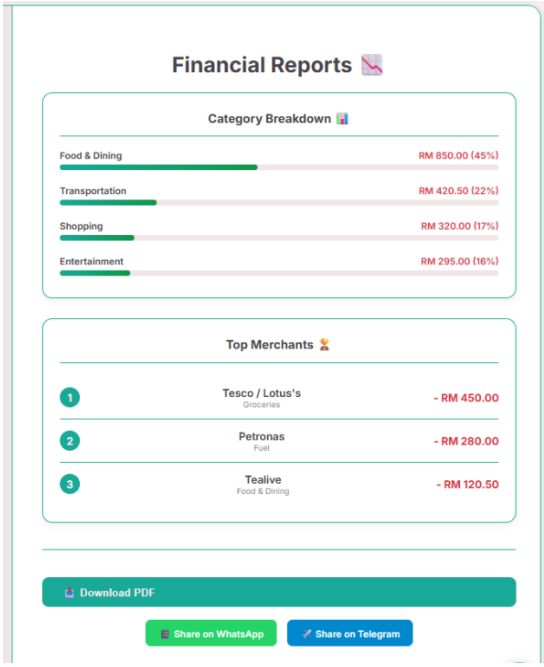


Figure 16 - Financial Reports

The Financial Tools section in Figure 15 provides access to three powerful features that help users take control of their financial future.

In Figure 16, The Reports & Analytics option opens the comprehensive reporting dashboard where users can:

- View category breakdowns showing spending distribution
- Analyze top merchants to identify where money is going most frequently
- Track monthly spending trends with comparisons to previous months
- Download reports as PDF for offline record-keeping or sharing with financial advisors
- Share reports directly via WhatsApp or Telegram with built-in social sharing buttons

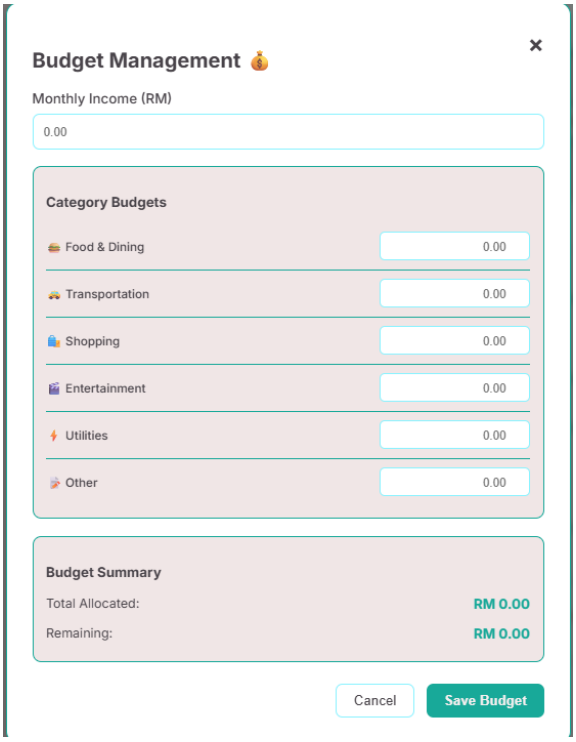


Figure 17 - Budget Management

The budget management option opens the comprehensive budgeting tool that enables:

- Monthly income setup to establish baseline for all budgeting decisions
- Category specific budget allocation for various expenses
- Real time calculation showing total allocated across all categories
- Remaining budget tracking with color-coded indicators (green for positive, red for negative)
- Validation to prevent negative income entries and ensure realistic budgeting
- Visual summary of how income is being distributed across spending categories

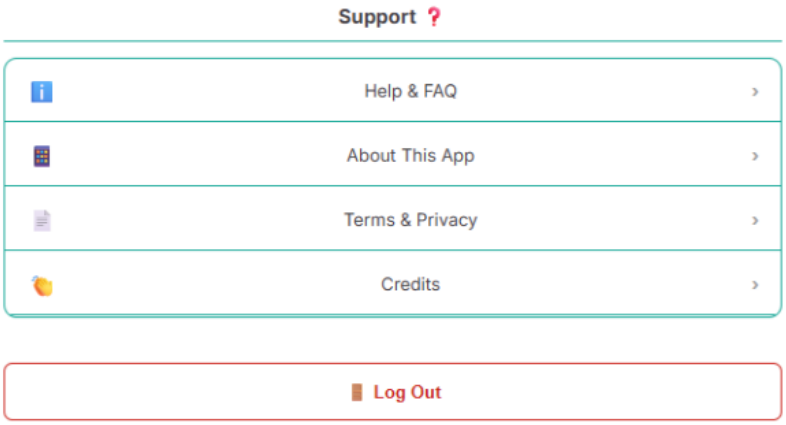


Figure 18 - Support Section

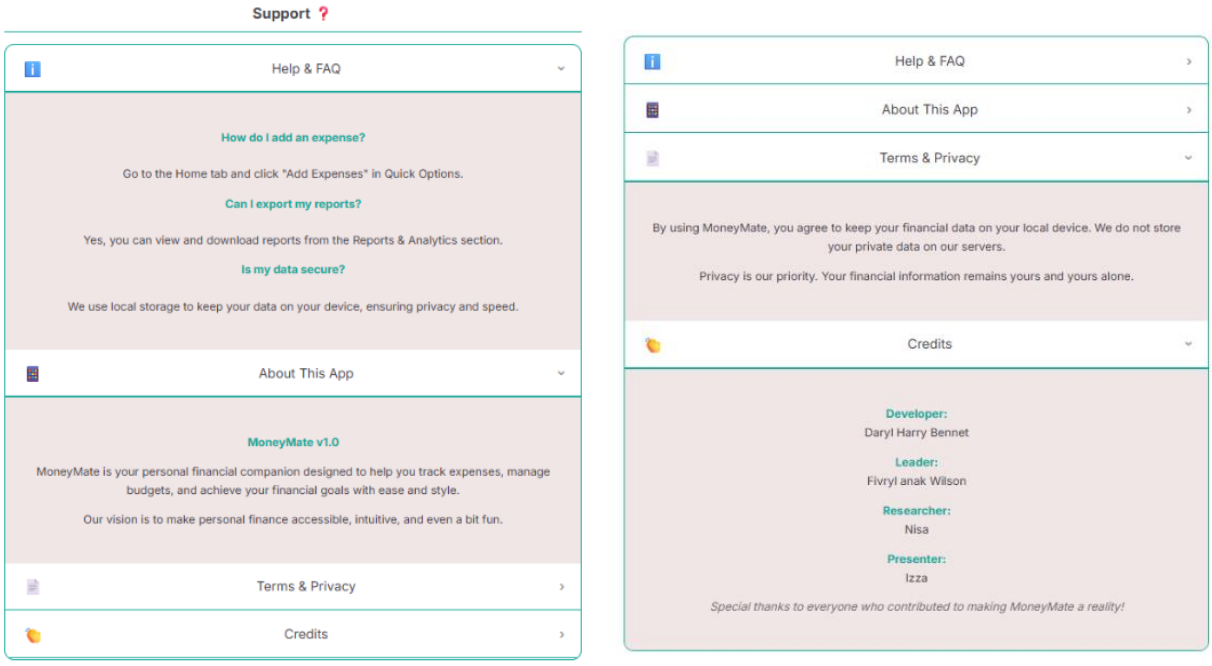


Figure 19 - Support section expanded

Support Section includes Help & FAQ, About this App, Terms & Privacy, and Credits with expandable content panels.

Logout Button is prominently displayed at the bottom for easy session termination.

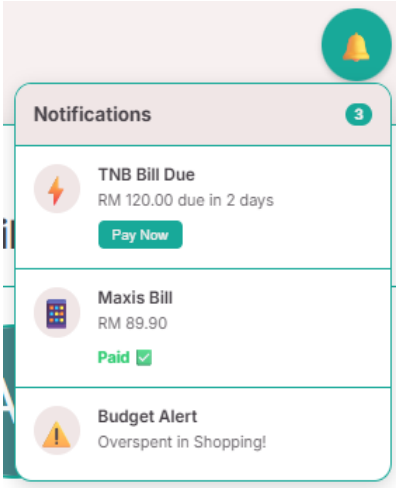


Figure 20 - Notification Dropdown

The notification system keeps users informed about important financial events and reminders. Clicking the bell icon in the header reveals a dropdown panel with prioritized alerts.

Notification Types:

- Warning (⚡): Overdue Bill reminder, includes a "Pay Now" action button for immediate response
- Success (📱): Paid status indicator showing completed transactions
- Alert (⚠): For Budget Alert as warning when category limits are exceeded

The notification header displays the count of unread alerts (3) and each notification uses color-coded icons and status indicators to help users quickly prioritize their attention. The dropdown animates smoothly and closes when clicking outside, providing a polished user experience.

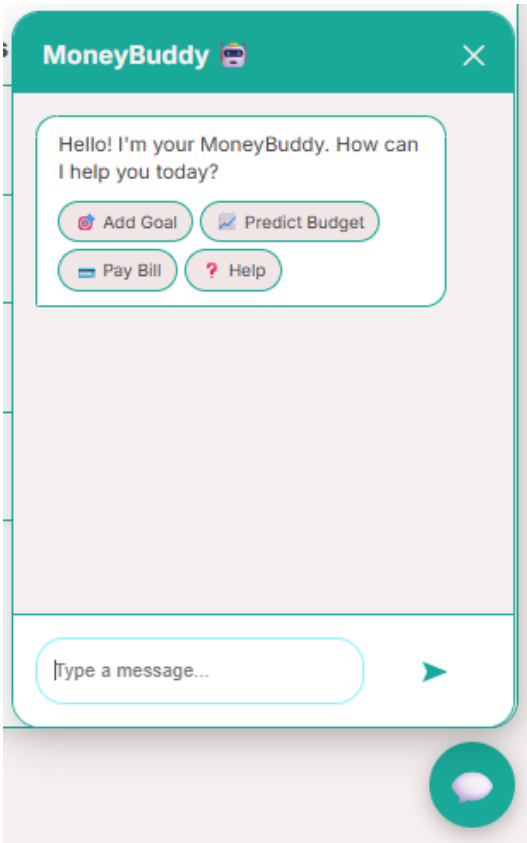


Figure 21 - AI Chatbot: MoneyBuddy

The MoneyBuddy chatbot provides conversational assistance for common financial tasks directly within the app.

Add Goal Flow: Users can create new savings goals through conversation. The bot prompts for goal name and target amount, then automatically adds a new goal card with progress tracking to the dashboard

Budget Prediction: Based on the user's current monthly spending (RM 3,200), the bot calculates and displays a 3-month spending forecast along with remaining balance projections.

Payment Assistance: Users can select from available bills and initiate payments directly through the chat interface, which triggers the payment modal with pre-filled details.

The chatbot is designed to make financial management more accessible through natural conversation, reducing the learning curve for new users and providing quick access to core features.

MoneyMate Dark Mode

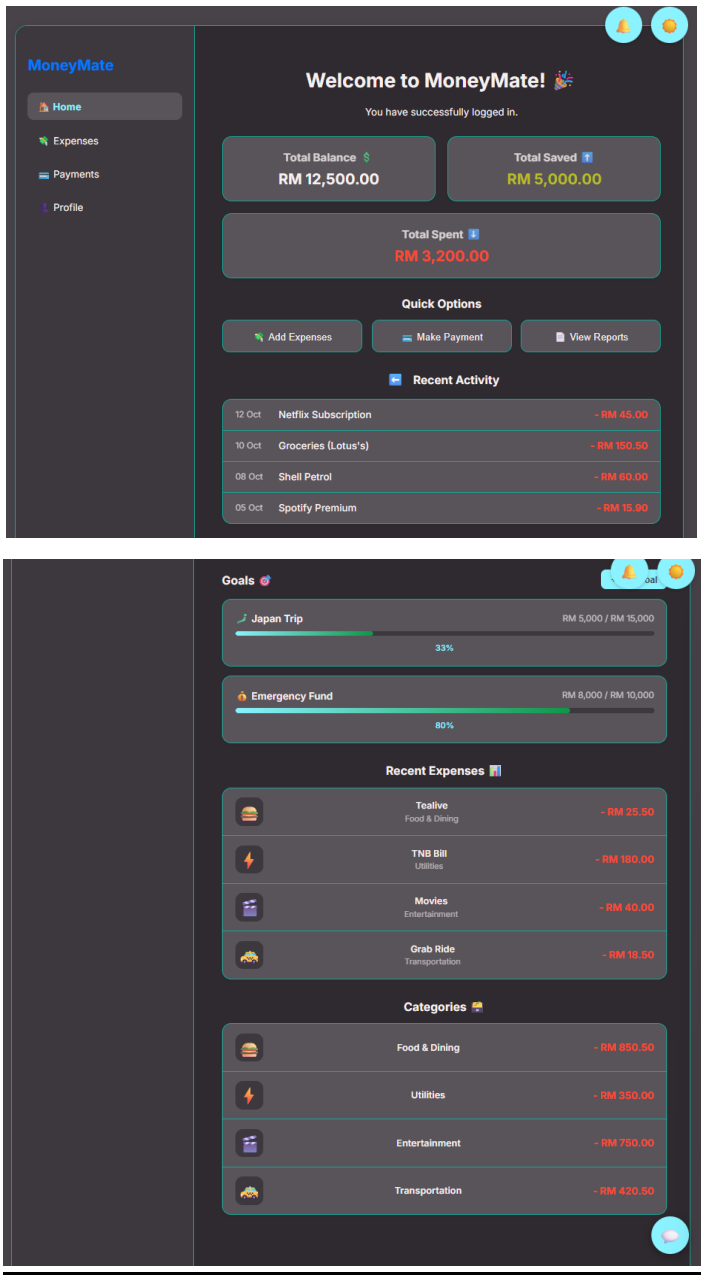


Figure 22 – MoneyMate Dashboard in dark mode

All interfaces support dark mode with a single click of the / button in the header. The theme persists across sessions using local storage and applies consistent colour adjustments throughout the application, thus transforming backgrounds, text, and accent colours for comfortable nighttime viewing. This small but meaningful feature reduces eye strain and improves accessibility for users who prefer darker interfaces.

AI ACKNOWLEDGEMENT

Disclosure

In compliance with competition requirements, we disclose all artificial intelligence tools used during the development of MoneyMate. Failure to disclose AI usage may result in disqualification, and we affirm our commitment to transparency throughout this project.

AI Tools Used

Tool	Purpose	Usage Explanation
Google Gemini API	Assisted with code generation, debugging, and problem-solving during development	Used extensively throughout development for code generation, troubleshooting errors, and refining implementation. When the developer encountered issues or needed to generate specific code snippets, Gemini was consulted to provide solutions and alternatives
Antigravity	AI-Integrated development environment used similarly to VSCode with AI assistance throughout the coding process	Served as the primary AI-powered development environment, functioning similarly to VSCode but with integrated AI assistance. This tool helped streamline the coding process by offering real-time suggestions and automating repetitive tasks

Chatbot Implementation

The MoneyBuddy chatbot feature is built using hardcoded JavaScript logic rather than external AI APIs. It simulates conversational interactions through:

- Predefined response patterns for goals, budget predictions, and payments
- Interactive buttons that trigger specific app functions
- Prompt-based goal creation flow
- Context-aware replies based on keyword matching

This approach provides a seamless user experience while maintaining reliable functionality within the prototype scope. The chatbot demonstrates how conversational interfaces can make financial management more accessible, even without complex backend AI systems.

Ethical Statement

All AI-generated code and content were reviewed, tested, and modified by human team members to ensure accuracy, security, and alignment with project goals. The team takes full responsibility for the final output and its implementation.

CONCLUSION

Project Summary

MoneyMate was developed to address a pressing challenge across ASEAN, which is the growing gap between access to digital financial services and actual financial literacy. While digital payments, e-wallets, and now pay later options become increasingly accessible, understanding of risk, interest, and debt management has not kept pace. MoneyMate tackles this problem by offering a centralized, user-friendly platform that goes beyond simple transaction tracking to prioritize financial education.

The prototype successfully demonstrates how technology can consolidate fragmented financial activities such as expense tracking, bill payments, goal setting, and budget management, into a single, intuitive interface. Key features such as the visual goal of progress tracking, and comprehensive reporting provide users with the tools they need to take control of their financial lives. The inclusion of dark mode and thoughtful UI design ensures accessibility and comfort across different usage scenarios.

Most importantly, MoneyMate lays out the groundwork for genuine AI-powered financial coaching. While full AI integration remains for future development, the foundation is set for personalized insights, proactive nudges, and educational content that can adapt to each user's unique financial journey.

Potential Future Improvements

- **Mobile Responsiveness:** Optimize the interface for mobile devices to ensure seamless financial management on the go, where most users interact with financial tools

- **Real Bank Integration:** Develop API connections with financial institutions to automatically import transactions, eliminating manual entry and providing real-time financial snapshots

- **Enhanced Notification System:** Build a fully functional notification engine that delivers timely reminders for bill payments, budget alerts, and goal milestones.

Final Thoughts

MoneyMate is more than just another budgeting app. It represents a step toward democratizing financial education across ASEAN, ensuring that as digital financial services continue to expand, the people using them are equipped with the knowledge and tools to do so wisely. The prototype proves the concept is viable, the design is intuitive, and the need is real. With continued development, MoneyMate has the potential to make financial literacy accessible, engaging, and genuinely life-changing for users across the region.